VICTORY 2027

Soil Resources

LECTURE-3 Geography

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Topics To Be Covered

1. Soil Erosion
2.
3.
4.

Discipline
Manners
Etiquette

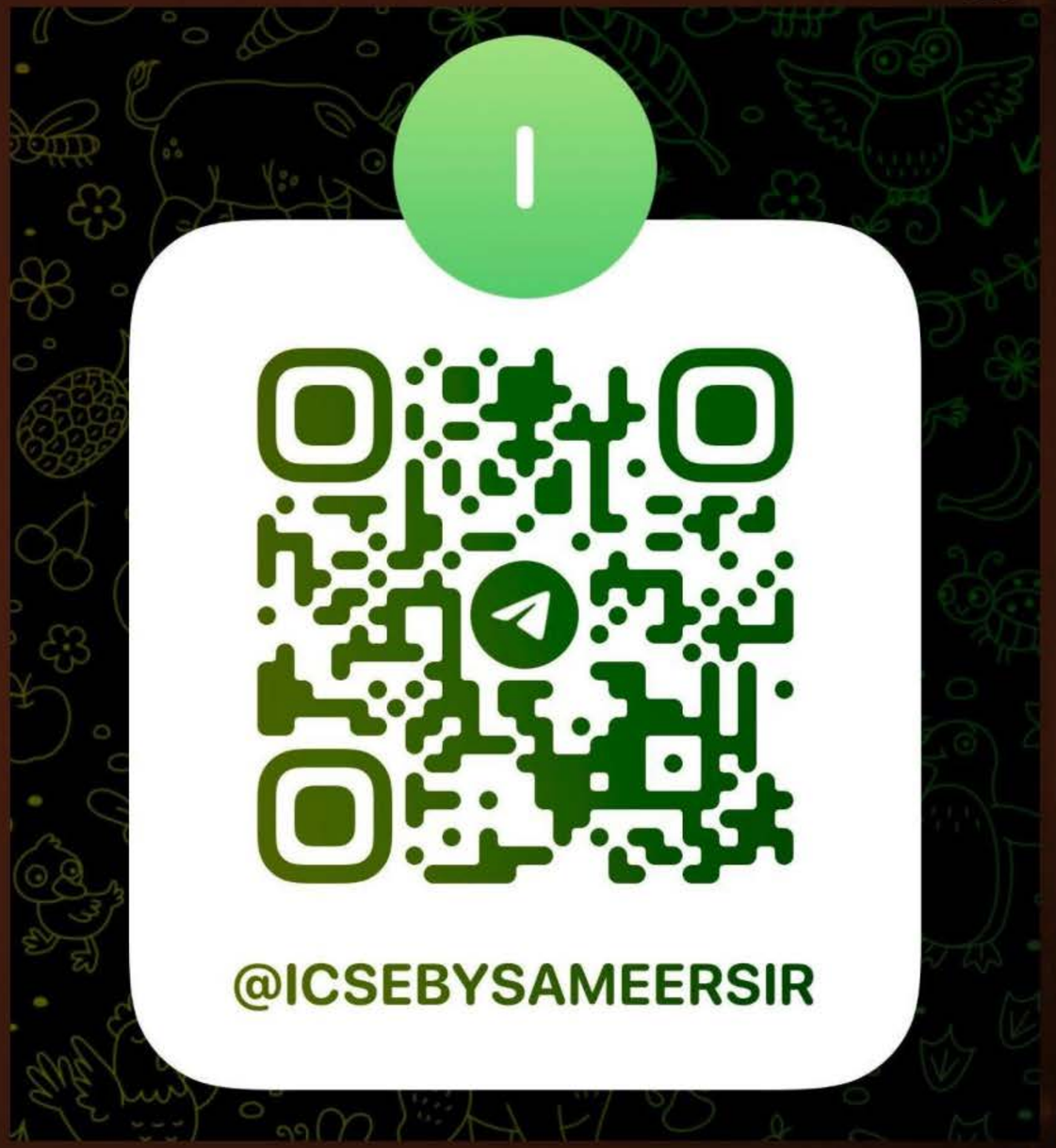
Soil Resources

- *Types of soil (alluvial, black, red and laterite), mode of formation, distribution, composition and characteristics such as colour, texture, minerals and crops associated.*
- *Soil Erosion – meaning, agents and causes.*
- *Soil Conservation – importance and methods*



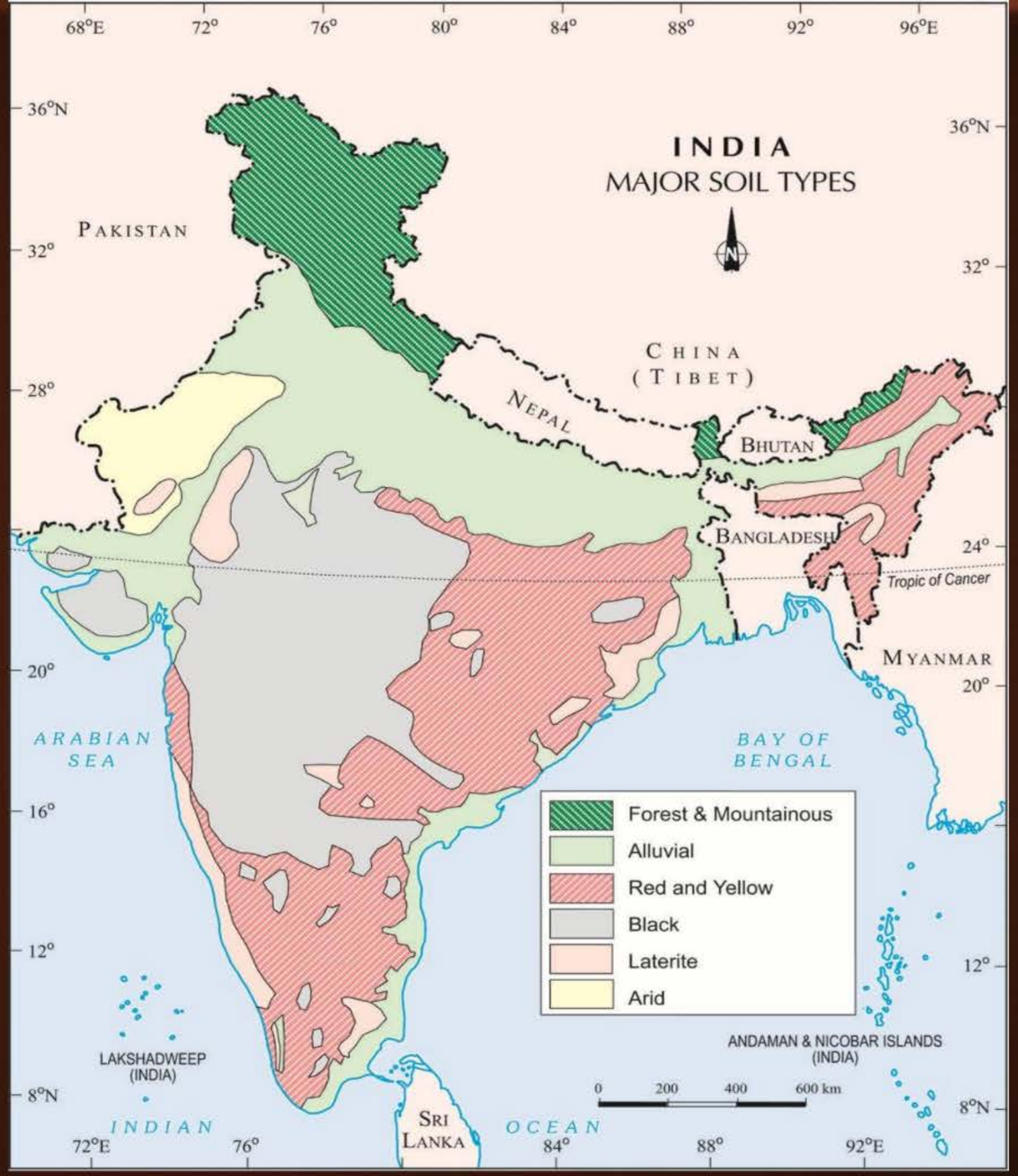
Telegram

https://t.me/icsebysameersir






Alluvial
Black
Red
Laterite


INDIA
MAJOR SOIL TYPES
N
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
Forest & Mountainous
Alluvial
Red and Yellow
Black
Laterite
Arid
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
SRI LANKA
0 200 400 600 km

SOIL EROSION

- **The removal of the topsoil cover by water, wind and human activities is called soil erosion.**
- Man is responsible for soil conditioning either through agriculture or through pollution like dumping of chemical wastes in water bodies as well as underground.
- Besides human activity, soils are degraded by nature during drought or floods.
  - सुखता
  - बाढ़

1. Soil Erosion by Water

(i) **Sheet Erosion:** It occurs on gentle slopes and is the slow removal of a thin layer of soil when vegetation is destroyed. Rainwater washes away the thin layer of bare soils.



1. Soil Erosion by Water

(ii) **Rill Erosion:** When sheet erosion continues for long, the silt-laden run-off forms many finger-shaped rills or grooves over a large area. This is called rill erosion. It is the intermediate stage between sheet erosion and gully erosion.

Grooves
नाले



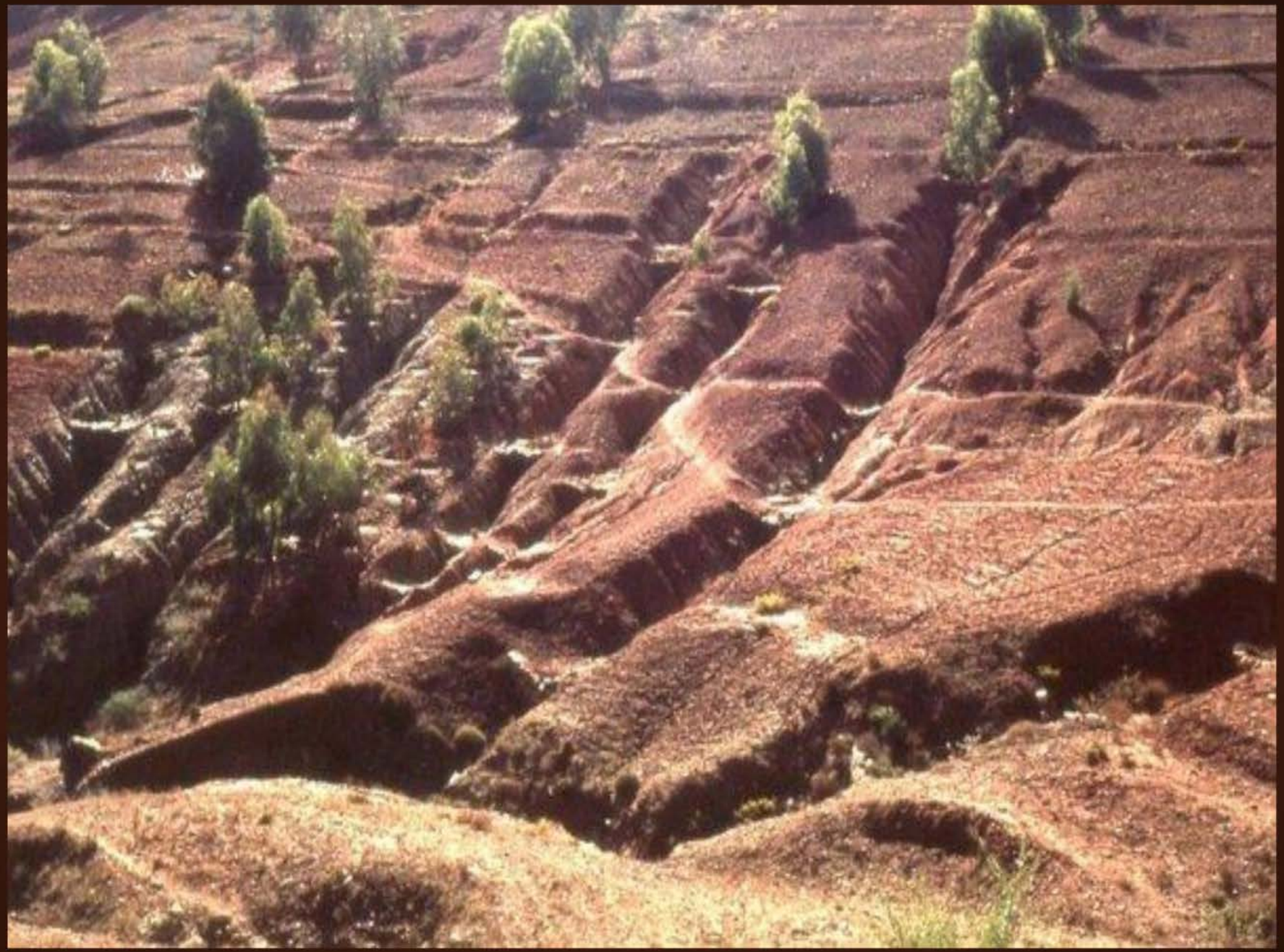
1. Soil Erosion by Water

(iii) **Gully Erosion:** During heavy downpour, deep gullies are made on bare soils on account of water run-off. Gully erosion removes nutrients and heavy load of loose soils, making the soil unproductive. It makes water very muddy. This is seen in the Chambal Valley region.



Fig. 1.11: Gully Erosion





SOIL EROSION

(iv) **Leaching:** After harvesting, farmers leave the soils bare for some time. During rainfall the nutrients in the soil are leached or percolated below the top layer.

किनार। लहरें

(v) **Sea or Shore Erosion:** The tidal waters of the sea cause considerable damage to the soil along the coast. The powerful waves dash against the coast and break hanging cliff rocks. The broken material is then removed by the retreating sea waves. This type of sea erosion is seen throughout the eastern and western coasts of India.



2. Soil Erosion due to Human Action

Cutting down of trees

- The loss of vegetation cover leads to soil Erosion on hilly slopes because water instead of sinking into the ground washes the soil down.



2. Soil Erosion due to Human Action

- The Outer Himalayas (the Shiwaliks), the Western and Eastern Ghats are subjected to deforestation by man.
- These areas also receive heavy monsoon rains.
- The removal of vegetation cover for different land use like railway lines, roads, buildings or even agriculture has caused Sheet, Rill and Gully erosion.
- In these areas, local population practise shifting cultivation.
- The heavy rains then wash away the bare soil from the slopes to the valleys below.
- Another important cause soil Erosion is uncontrolled grazing of domestic animals in the valleys and the upper slopes.

3. Erosion by Wind

- Wind Erosion refers to the movement and deposition of soil particles by wind.
- It occurs when soil devoid of vegetation is exposed to high-velocity wind. ↳Absent
- Wind moves soil particles 0.1-0.5 mm in size in bouncing or hopping fashion and those greater than 0.5 mm by rolling.
- The former is known as saltation and the latter as soil creep.

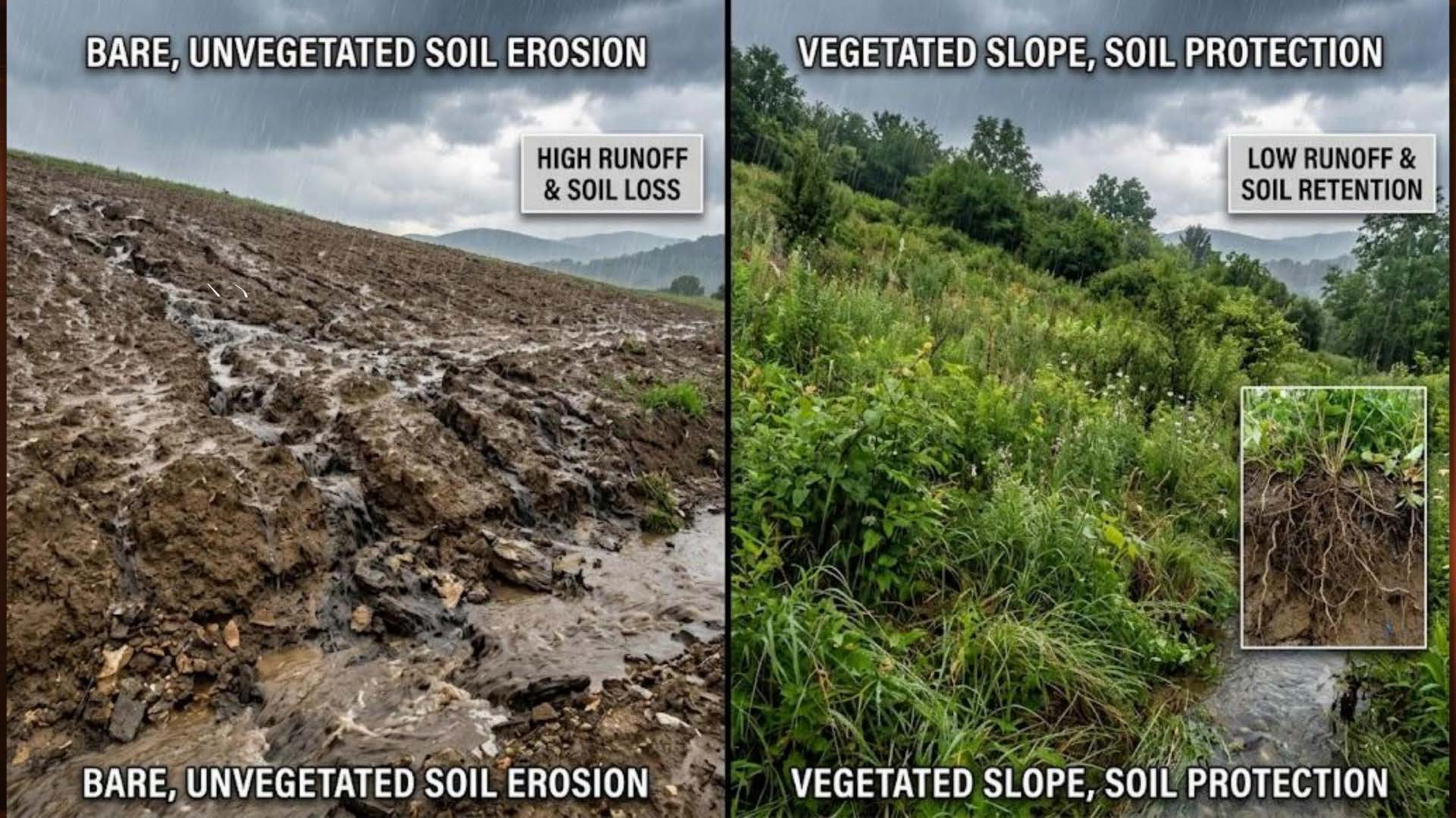
3. Erosion by Wind

- Wind erosion reduces the productive capacity of soil, as most of the nutrients required by the plants are carried away by the wind.

CAUSES OF SOIL EROSION

(i) **Heavy Population Pressure on Land:** India's forest cover continues to be very low, just * 21.54 per cent of the total area. On the other hand, the population continues to rise at a rapid rate and has already crossed the one billion mark in the year 2000. More forests are being destroyed to house and feed the increasing population. The heavy pressure on land is the main cause of soil erosion.

(ii) **Nature of Rainfall:** India receives 80 to 90 per cent of rainfall in the monsoon season. Heavy downpour during monsoon months causes floods. In the remaining months droughts are frequent. This affects the soils.


BARE, UNVEGETATED SOIL EROSION
HIGH RUNOFF & SOIL LOSS
VEGETATED SLOPE, SOIL PROTECTION
LOW RUNOFF & SOIL RETENTION
BARE, UNVEGETATED SOIL EROSION
VEGETATED SLOPE, SOIL PROTECTION



CAUSES OF SOIL EROSION

(iii) **Overgrazing:** The number of domestic animals particularly cattle in India is the highest in the world. Cattle freely graze in open lands making them bare of vegetation. Winds carry away dry soil particles from the bare landscape. Thus, soil erosion takes place. In many parts of Rajasthan, excessive grazing by cattle has resulted in the exposure of the topsoil to elements of denudation.



CAUSES OF SOIL EROSION

(iv) **Bad Farming Techniques:** The poor farmers plough fields in traditional ways. The farming techniques and small size of holdings lead to soil erosion on a large scale. The absence of terracing, contour cultivation, crop rotation and the improper use of manure etc., have caused serious problems of soil erosion.

(v) **Topography:** Northeastern parts of India, Shiwaliks and hilly regions in South India are affected by soil erosion because of steep slopes and heavy rainfall. During heavy rainfall, soils are washed away by running water down the slope.

(vi) **Deforestation:** Forests are destroyed so that more land can be used for cultivation. Cutting of trees exposes the soil to water and wind, which leads to soil erosion.





CAUSES OF SOIL EROSION

Regions of Soil Erosion

- In India, soil erosion is one of the great enemies of Indian agriculture responsible for low agricultural productivity.
- According to an estimate every year rainwater alone washes out 1.25 mm thick fertile topsoil.
- Among the States, Rajasthan comes on the top of the soil eroded regions, followed by Madhya Pradesh, Maharashtra, Uttar Pradesh, Gujarat, Andhra Pradesh and Karnataka.



SAVE THE SOIL

It takes ten thousand years for soil to form and merely ten years to destroy. Globally 52 per cent of agricultural land is already degraded. Poor soil leads to poor nutritional value. Today's fruits and vegetables already contain 90 per cent fewer nutrients. Depleted soils cannot absorb and regulate water flows. Lack of water retention leads to water scarcity, droughts and floods. According to scientists, 27,000 species of life forms are becoming extinct every year due to loss of habitat. Loss of biodiversity further disrupts the soil habitat and prevents soil regeneration. Another alarming fact is that soil stores three times more carbon than living plants and two times more than the atmosphere.

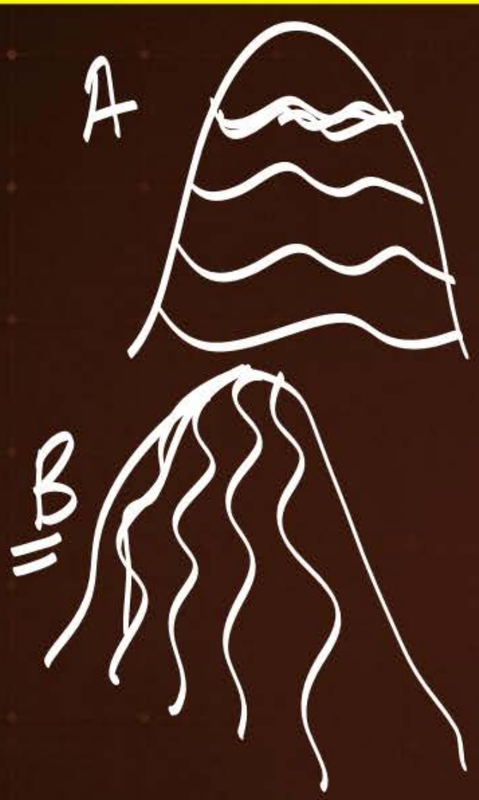
The Save Soil Movement is a global campaign focused on addressing the critical issue of soil degradation with the primary goal of increasing the organic content in cultivable soil to ensure its long-term health and productivity. The movement began with the aim to inspire 3.5 billion people worldwide, especially national leaders and policy makers, to support policies that enhance soil health. Various campaigns and initiatives are conducted to raise awareness about the importance of soil health because improving soil health would enable us to combat climate change, enhance biodiversity, and ensure food security.

Soil Conservation

1. **Terrace Farming:** On hilly slopes, Terraces are levelled into step-like small fields with even slope and prevent the soil from being washed away by running water. Terrace farming is practised with successful results in Japan, South-East Asia and the USA.



Soil Conservation



2. **Shelter Belts:** Farmers plant trees in several rows to check wind erosion. They are known as wind breaks.

3. **Contour Ploughing:** Ploughing along contours on a slope prevents soil being washed away by rainwater or by surface run off. Contours act like bunds. Contour ploughing is common in Japan, China and some South-East Asian countries.





TRADITIONAL UP-AND-DOWN PLOUGHING



GULLY EROSION WITH STRAIGHT FURROWS

UNPROTECTED: HIGH SEDIMENT RUNOFF

CONTOUR PLOUGHING



SOIL RETAINED & INFILTRATION INCREASED

PROTECTED: REDUCED RUNOFF & SOIL RETENTION

Soil Conservation



4. **Strip Cropping:** Crops are grown in alternate strips of land to check soil erosion by water or wind. The crops like hay and wheat are alternated with strip of crops such as corn, soya beans and cotton. This method helps to prevent soil erosion by having the roots of crops hold on to the soil and prevent it from being washed away by water or wind.

5. **Construction of Dams:** Rivers cause soil erosion. Dams are built in the upper course of rivers to control erosion of soil.



Soil Conservation

6. **Plugging Gullies:** The gullies made in the soil are plugged with deposition of silt during heavy rains.

7. **Planting Trees:** The trees, like in the case of Shelter Belts, are planted along the edges of the fields, the waste land and on steep slopes to prevent soil erosion as well as to enhance the capacity of the soil to retain water.

Soil Conservation

- Soil conservation means prevention of soil loss from erosion or prevention of reduced fertility of soil caused by over usage, acidification, salinisation or ther types of soil contamination.
- There is an acute need to prevent soil erosion, because of its adverse effects which include the following:

Soil Conservation

(i) Loss of fertile topsoil together with its mineral nutrients, from the upper surface lead to gradual loss of soil fertility and agricultural productivity.

(ii) Lowering of the water table and decreasing soil moisture.

(iii) Drying of vegetation and extension of arid lands (dry land) leading to an increase in the frequency of droughts and floods.

(iv) Silting of river and canal beds.

(v) Recurrence of landslides.

(vi) Adverse effect on economic prosperity and cultural development.


Erosion
Water
Sheet
Rill
Gully
Leaching
Wind
Sea shore
Human
Deforestration
Bad agricultural practices
Increasing population
Construction activities
Pollution
Overgrazing
Measures
Terrace farming
Shelter Belts
Contour Ploughing
Constructing dams
Afforestation
Strip Cropping
Thank You